

Vikas Kumar

Indian Institute of Information Technology Bhagalpur

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EDUCATION

• Indian Institute of Information Technology, Bhagalpur

B.tech in Mechatronics Engineering

November 2022 – May 2026

CGPA: 7.4

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, C, HTML, CSS

Frontend: React.js, Tailwind CSS, Responsive UI Design

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Access Control (RBAC), Redis, Caching

Databases: MongoDB, PostgreSQL, Firebase

Cloud & DevOps: AWS, Postman, Kubernetes, Docker, Git, GitHub, Vercel, Render

Developer Tools: VS Code, Jupyter Notebook

Core CS Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Operating Systems, Computer Networks, Database Management Systems (DBMS)

Areas of Interest: Full Stack Web Development (MERN/MEAN), Machine Learning

EXPERIENCE

• CyberWarFare Labs

Full Stack Engineer Intern

June 2026 – Present

Bengaluru, Karnataka

- Enhanced the **CWL's CTF platform's user & admin panel** with **responsive UI components** using **Bootstrap** and **React.js**, improving **usability** and **UI consistency** across **10+ core modules**, reducing user-reported UI issues by **30%**.
- **Developed and deployed 15+ REST APIs** using the **MERN Stack** with **JWT-based authentication** and **Role-Based Access Control (RBAC)**, validating **20+ API workflows** via **Postman** for reliable and secure **frontend-backend integration**.
- Configured and integrated **SMTP-based email notification system** enabling automated **transactional emails** (account verification, password reset, challenge alerts) for **1000+ platform users**, improving **communication reliability** and **user engagement**.
- Implemented **Artifact** and **Lab Challenge** modules with isolated **VPN** and **Kubernetes-based** lab environments, enabling **scalable hands-on cybersecurity training** infrastructure supporting **concurrent multi-user access**.

PROJECTS

• QuickStay | React.js, Node.js, MongoDB, Clerk Auth, Tailwind CSS

[GitHub](#) / [Live](#)

June 2025

- Engineered **QuickStay**, a full-stack hotel platform, architecting features for real-time search and secure bookings across **500+ hotel listings**.
- Optimized frontend-backend communication using **React.js**, **Node.js**, and **MongoDB**, reducing API response time by **35%** through indexing and query optimization.
- Engineered adaptive pricing engine and intelligent room filters, cutting irrelevant search results by **50%** and improving booking conversion accuracy by **25%**.

• SupportAgentAI | Python, Scikit-learn, TF-IDF, Logistic Regression, Pandas

[GitHub](#)

September 2026

- Engineered an end-to-end **customer-support automation system** that classifies intents, retrieves historical responses, detects risk, and routes requests for **auto-handling or human review**.
- Processed a **2.8M+ tweet** support dataset into **149,680** cleaned customer-support pairs and built a **TF-IDF + Logistic Regression** pipeline across **10 intent classes**.
- Implemented **TF-IDF cosine-similarity retrieval** with a **200,000-feature** index and leakage-free evaluation, achieving **49.00% intent accuracy** and **49.75% Macro F1** on a 200-example human-gold benchmark.
- Developed deterministic **risk-based escalation** using confidence/retrieval thresholds, security/payment/legal rules, and automated failure analysis; validated the pipeline with **100% escalation recall** and **10/10 regression tests passed**.

CERTIFICATIONS

• The Complete Web Development Bootcamp – Udemy, Angela Yu (2025)

[View Certificate](#)

• AWS Cloud Practitioner Essentials – AWS Training & Certification (2026)

[View Certificate](#)

ACHIEVEMENTS

• Smart India Hackathon 2024 – Selected in the inter-college round out of 200+ teams, presented an AI-based solution to a real-world government problem statement.

[View Certificate](#)

• Competitive Programming – Solved **600+ DSA problems** across LeetCode, CodeChef, and Codeforces, maintained a **450+ days** continuous streak on CodeChef and **150+ days** streak on LeetCode.

• Carrom Champion – Secured **1st position** in the Team Carrom event at Udyatva Inter-College Sports Fest 2025.